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Oppg 1/

$$E[Y \mid X = x] = \beta_0 + \beta_1 x \qquad Var[Y \mid X = x] = \sigma^2$$

$$\bar{y} = 67.7 \qquad \bar{x} = 42{,}49 \qquad \sum_i^{n=25} (x_i - \bar{x})^2 = 14.29 \qquad \sum_i (x_i - \bar{x})(y_i - \bar{y}) = 18.84$$

$$SSE = \sum_{i=1}^{n} \left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x)\right)^2$$

$$\frac{\partial SSE}{\partial \hat{\beta}_0} = -2\left[\sum y_i - n\hat{\beta}_0 - \hat{\beta}_0 \sum x_i\right]$$

$$\frac{\partial SSE}{\partial \hat{\beta}_1} = -2\left[\sum y_i x_i - \hat{\beta}_0 \sum x_i - \hat{\beta}_1 \sum x_i^2\right]$$

$$\left.\begin{array}{l} \frac{\partial SSE}{\partial \hat{\beta}_0} = 0 \\ \frac{\partial SSE}{\partial \hat{\beta}_1} = 0 \end{array}\right\} \Rightarrow$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 67.7 - 1.3184 \cdot 42{,}49 = 11{,}6811$$

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x}) y_i}{\sum (x_i - \bar{x})^2} = \frac{18.84}{14.29} = \underline{1{,}3184}$$

Oppg 2/

$$n = 20 \qquad s_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = -1211 \qquad \bar{x} = 26.18$$

$$s_{xx} = \sum (x_i - \bar{x})^2 = 2738 \qquad \bar{y} = 72.99$$

$$s_{yy} = \sum (y_i - \bar{y})^2 = 1179$$

$$\varepsilon_i \sim N(0, \sigma^2) \qquad i \in [0, n]$$

$$Y_i = \alpha + \beta x_i + \varepsilon_i \qquad Y \sim N(\alpha + \beta x_i, \sigma^2)$$

a)

$$SME(\beta) = \hat{\beta} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{-1211}{2738} = \underline{\underline{-0{,}44229}}$$

b)

$$SME(\alpha) = \bar{y} - \hat{\beta}\bar{x} = 72.99 - (-0.44229) \cdot 26.18 = \underline{\underline{84{,}569}}$$

α = konstantledd

fordi $\varepsilon_i \approx$ feil i regresjonsmodell (min tolkning)

c)

$$SSE = \sum \left(y_i - (\hat{\alpha} + \hat{\beta} x_i)\right)^2 = 643$$

$$\frac{1}{\sigma^2} SSE = n \cancel{\Rightarrow}$$ Ikke forventningsrett

si $\hat{\sigma}^2 = \frac{SSE}{n}$ og $\frac{n\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-2}$

$$E[\hat{\sigma}^2] = \frac{n-2}{n}\sigma^2 \Rightarrow s^2 = \frac{n}{n-2}\hat{\sigma}^2 = \frac{n}{n-2} \cdot \frac{SSE}{n} = \frac{643}{18} = \underline{\underline{35{,}722}}$$

d) $\mathrm{Var}[\hat{\beta}] = \dfrac{\sigma^2}{\sum (x_i - \bar{x})^2} \Rightarrow \widehat{\mathrm{Var}[\hat{\beta}]} = \dfrac{s^2}{2738} = \dfrac{35{,}722}{2738}$

$$\widehat{SD(\hat{\beta})} = \sqrt{}$$

$$= \underline{\underline{0{,}11422}}$$

e) $\mathrm{Var}[\hat{\alpha}] = \sigma^2 \cdot \dfrac{\sum x_i^2}{\sum (x_i - \bar{x})^2} \approx s^2 \left[\dfrac{1}{n} + \dfrac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right] = 35{,}722 \left[\dfrac{1}{20} + \dfrac{26{,}18^2}{2738} \right] = \underline{10{,}728}$

$$\widehat{SD[\hat{\alpha}]} = \sqrt{10{,}728} = \underline{\underline{3{,}2754}}$$

f) $1 - \dfrac{SSE}{S_{yy}} = R^2$

$$= 1 - \frac{643}{1179} = \underline{\underline{0{,}45462}}$$

g) $s_x = \widehat{SD[x]}$

$S_{xx} = \sum (x_i - \underset{\uparrow \atop E[x]}{\bar{x}})^2 = \sum (x_i - \bar{x})^2$

definisjon til Varians[X]:

$\mathrm{Var}[X] = E[(X - E[X])^2]$

$S_{xx} = \sum \mathrm{Var}[X] = (n-1) s_x^2 \longleftarrow$ ikke $n s_x^2$ fordi $\sum_{i=1}^{n} (x_i - \bar{x}) = 0$

$$s_x = \sqrt{\frac{S_{xx}}{n-1}} = \sqrt{\frac{2738}{19}} = \underline{\underline{12{,}004}}$$

h) $s_y = \sqrt{\dfrac{S_{yy}}{n-1}} = \underline{\underline{7{,}8773}}$

j) $\mathrm{Cov}(x,y) = -63{,}737 \Rightarrow \left(\dfrac{\mathrm{Cov}(x,y)}{s_x s_y} \right)^2 = \dfrac{\left(\frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y}) \right)^2}{\frac{S_{xx}}{n-1} \cdot \frac{S_{yy}}{n-1}} = \dfrac{\left[\sum (x_i - \bar{x})(y_i - \bar{y}) \right]^2}{\left[\sum (x_i - \bar{x})^2 \right] \cdot \sum (y_i - \bar{y})^2} = \dfrac{\mathrm{Cov}(x,y)^2}{\mathrm{Var}[X]\,\mathrm{Var}[Y]} = \left[\mathrm{Corr}(x,y) \right]^2$

$$\left(\mathrm{Corr}[x,y] \right)^2 = R^2 = \underline{\underline{0{,}45462}}$$